

MOSAIC Recommendations UI - Complete Guide

Product: MOSAIC Intelligence (AI Tools)

Project: tsp-mosaic-frontend | Local: http://localhost:3010

1. Overview

Campaign-centric dashboard for optimization suggestions from MOSAIC, Google Ads, and Meta. The dashboard lists campaigns only. Campaign detail shows full recommendations by source, hierarchy, compare, and history.

2. Routes

- /recommendations/all - Campaign dashboard (hub)
- /recommendations/[campaignId] - Single campaign detail
- /recommendations/compare - All campaigns compare workspace
- /recommendations/compare/[campaignId] - One campaign compare

3. Dashboard

KPI cards

- Total, Applied, Dismissed, Ignored, Pending, Avg lift

Campaign table

- Campaign: name, NEW badge, status, platform
- MOSAIC, Google, Meta: recommendation counts per source
- Pending, Avg lift, Schedule (frequency + last run)
- Open: navigate to campaign detail
- Compare: shown when 2+ sources have recommendations
- Filters: search, status, sort by pending or lift
- Row hover prefetches campaign route for faster open

4. Campaign Detail

Header

- Campaign info, budget, bid strategy, source count chips

Tab: By source

- Source tabs: All, MOSAIC, Google, Meta
- PRIMARY ACTION: main campaign-level recommendation
- SUPPORTING ACTIONS: nested by ad set and segment
- Campaign-level supporting and orphan recommendations
- Filter by module: Budget, Bid, Audience, Placement, Geo/Daypart, Dismissed
- Info banner when other sources also have suggestions

Tab: Cross-platform

- Side-by-side MOSAIC vs Google vs Meta (ComparePairPanel)
- Match types: aligned, similar, conflict, mosaic_only, platform_only

Tab: History

- Audit log: applied, dismissed, rolled back with actor and timestamp

Actions

- MOSAIC: Apply (with note), Dismiss, Undo (24h), bulk select
- Google/Meta: view-only, link to Open in ad manager
- No checkbox on platform-native cards

5. Compare Workspace

- Compare hub lists all campaigns with alignment summary
- Campaign compare shows pairs by category (budget, bid, etc.)

6. Components

- AppHeader - top navigation
- CampaignSummaryTable - dashboard table
- PrimaryRecommendationCard - main action card
- RecommendationCard - supporting recommendation
- RecommendationHierarchy - ad set / segment nesting
- RecommendationScoreBadges - lift, confidence, risk
- SourceTag - MOSAIC, Google, Meta badge
- ApplyConfirmModal - confirm before apply
- BulkActionBar - bulk apply for selected MOSAIC recs
- CompareSummaryTable - compare hub table
- ComparePairPanel - side-by-side compare cards
- HistoryPanel - audit timeline

7. Data Layer

- mocks/recommendations.ts - MOSAIC demo with parent/child links
- mocks/platformRecommendations.ts - Google and Meta demo
- lib/allRecommendations.ts - merges all sources
- lib/campaignSourceCounts.ts - per-campaign source counts
- lib/recommendationTree.ts - primary to ad set to segment tree
- lib/recommendationCompare.ts - cross-source alignment
- lib/campaignDetailCache.ts - instant mock data for fast load
- API /api/recommendations/* - mock endpoints and mutations

8. Hooks

- useRecommendations - dashboard data
- useCampaignRecommendations - campaign + recs + compare pairs
- useCampaignHistory - audit log
- useApplyRecommendation, useDismiss, useUndo, useBulkApply - mutations

9. Recommendation Modules

- budget_orchestration -> Budget (raise budget, reallocate)
- bid_policy -> Bid (bid cap, target CPA)
- audience_strategy -> Audience (overlap, lookalike)
- placement -> Placement (Reels, publishers)
- geo_daypart -> Geo and Daypart (geo shift, dayparting)

10. User Flow

Dashboard -> Open campaign -> By source (apply MOSAIC) -> Cross-platform tab -> Compare button or header
Compare -> full workspace.

11. Performance

- Use port 3010 (npm run dev or npm run dev:clean)
- First campaign open in dev compiles JS once (~10-25s)
- loading.tsx shows skeleton during compile
- Lazy-loaded History and Compare tabs
- localhost:3000 is SpurAdConnect, not MOSAIC

12. Live Integration

Google and Meta APIs require a backend mapper to the normalized Recommendation shape. UI is ready: platform recs stay flat, compare uses category heuristics, Apply for platform recs via their APIs or Open in manager links.